SPRINT 8 · RESEARCH & MODEL STRATEGY · COMPANION TO THE BUILD INSTRUCTION

One model, many tables, no oracle

This document answers the architecture questions directly: which capability runs on which machinery, what happens when the world changes under us, how we benchmark without fooling ourselves, and how Rafeh's Model Master Rules document is actually used. Every claim traces to a verified paper (page 9) or a file in our own repo.

Do virality, best times and flop detection share one model?

No, and that is deliberate. ONE trained model exists in Sprint 8: a LightGBM gradient-boosted tree model that scores drafts against the page's own ladder. Flop detection is that model's calibrated output, not a separate system. Best Times (S9) is NOT a model prediction: it is the page's own history aggregated into slots, with Prophet reading the weekly rhythm. "Virality" has **no model at all, ever**: the top-10% tag is grading of the answer sheet, never a forecast. Full map on page 2.

What if trends change in the future?

Handled in four layers: monthly **intervention-aware retraining** (the model relearns culture), the **drift watchdog** (catches silent rot without waiting for labels), **perishable timing tables** (weekly refresh, drift-forced rebuild), and **era discipline** (a metric-definition break is never confused with a taste change). Pages 5-6.

What if we cannot get the topic trend score?

Ship without it: the research says the loss is small. The one study that added external trend signals (Wikipedia pageviews) measured only **+0.015 AUC-PR** over text-only. The World feature group is optional by design: v1 trains without it, the trend badge defers to fast-follow, and nothing else depends on it. Page 4.

How does Rafeh's rules document help?

It is the **constitution**: 60+ numbered, citable rules (MR-x.y) with honesty tags saying which are confirmed vs placeholders. Tickets and QA gates cite rule ids instead of re-arguing decisions; its conflicts table forced the target redefinition into the open. Page 8.

VIRALDATA / ARTEMISAI · 28 JUL 2026 · SOURCES: 58 VERIFIED PAPERS · THE MARVEL REPO AUDIT · MODEL MASTER RULES 24 JUL · VD-015B FINDINGS · METRIC-DEPRECATION ESCALATION 25 JUL

01 The model map: every capability, its machinery, and why

CAPABILITY	MACHINERY	TRAINED MODEL?	WHY THIS CHOICE (EVIDENCE)
Draft flop warning (S8)	**LightGBM** gradient-boosted trees, ONE global model, page-relative percentile label + isotonic calibrator + TreeSHAP reasons	**YES · the one model**	Medium tabular data with heavy tails is exactly where trees beat deep learning; every SMP Challenge podium 2023-2025 is GBDT over frozen embeddings
Band + p_flop + reasons	The SAME model's output, calibrated (isotonic) and explained (SHAP)	**same model**	Flop detection is not a second system; one source of truth `MR-1.5`
Draft Duel (rank my 2-3 variants)	The SAME model used pairwise: score difference within one page cancels shared luck	**same model**	Comparison formally beats absolute scoring when noise is high (Shah 2014); config change, not a new stack
Upside odds + pattern tables	Per-page aggregate tables: format × topic × day × hour → average spread, odds, sample count. Monthly refresh, 5-sample floor	**NO · lookups**	Historical frequencies are unfalsifiable and explainable; a model would add error to something counting answers exactly `MR-2.4`
Best Times (S9)	Per-page 24h × 7d slot aggregation (normalised, shrunk) + Prophet weekly seasonality per page + changepoint drift detection	**NO LightGBM** · Prophet fits per page	Timing is a property of the audience's rhythm, not of a draft; different data grain, different refresh cadence (weekly vs monthly) `MR-7.2` `MR-7.3`
"Virality" / top-10% tag	The answer sheet: historical grading on the page ladder. Applied AFTER outcomes exist	**NO MODEL, EVER**	Winner-side prediction is dead: AUC 0.51 on our data, and the ceiling is physics (models explain under half of success even with unlimited data) `MR-1.1`
Day-2 maturity grader (S9 spec)	Tiny per-page log-linear scaler (day-2 → day-30), upgraded with daily-delta features	**separate tiny model**	Different job (post-publish forecasting from early observation); canonical Szabo-Huberman line of work, honest at daily grain
Model-health watchdog (S9)	Statistical detectors on live outputs: KS tests on confidence distributions + no-label accuracy estimation, joint alert rule	**NO · statistics**	Labels never arrive in production; detectors on the models' own outputs are the field's best answer (Failing Loudly)
Art-E (S8)	Claude-class LLM + typed query-template tools + citation gate	**NO training** · frontier API	The LLM narrates; every number is computed by a pre-tested query. Free-form SQL benchmarks at ~21-82%: disqualifying
FUSION content type (exists, S6)	CLIP + small MLP (Macro F1 0.93) + Haiku vision labels	**exists**	Feeds the picture features into the LightGBM; not rebuilt in S8

THE ANALOGY: A KITCHEN HAS ONE HEAD CHEF, NOT ONE MACHINE

The **chef** (LightGBM) tastes a dish before it goes out and says "this risks disappointing, here is why". The **recipe book** (pattern tables) just records what worked before: nobody predicts a recipe book, you look things up in it. The **maître d'** (Prophet) knows when the room fills up: a rhythm, not a prophecy. The **food critic's yearly list** (top-10% tag) is written after the meals happened. And the **waiter** (Art-E) explains any of it, reading from receipts. One mega-model doing all four jobs would entangle their failure modes and make every mistake harder to find.

Why exactly one trained scoring model: (1) pages are too small to train alone, so page segregation lives in the label and features, not in per-page models `MR-2.1`; (2) the targets differ in kind: draft scoring predicts an unpublished thing, timing describes audience rhythm, grading records outcomes: forcing one model onto all three would blur what each is allowed to claim; (3) accountability: when the flop warning is wrong we retrain one model, when a heatmap misleads we fix one aggregation, and neither breaks the other.

02 The performance model, precisely

COMPONENT	CHOICE
Algorithm	**LightGBM** (gradient-boosted decision trees), regression objective on the ladder percentile; pinned v4.7; no linear_tree (breaks SHAP)
Label (D0 pending)	Within-page trailing-12-month percentile of spread. Candidate A: log(shares) percentile + absolute floor. Candidate B: shares per viewer. Era-flagged either way
Features	Picture: CLIP ~100 PCA dims + FUSION type. Words: caption stats (+ NLP labels as free passengers). Timing: hour, weekday, gap. Page state: rolling median, trend slope, frequency (as-of-date). World: OPTIONAL (page 4)
Calibration	Isotonic on P(bottom band), fit with clip at bounds, global (needs ~1,000+ points; never per page), refit after any target or era change
Reasons	TreeSHAP per prediction; ~100 CLIP dims summed into one "image content" reason (additivity makes grouped sums valid); raw log-odds phrased as "pushing up / down", never percentage points
Retrain	Monthly, intervention-aware (the log records what advice was shown); pattern tables monthly; timing tables weekly

Never inputs: reach, viral reach, likes, shares, comments, clicks, or anything derived. They do not exist at draft time; they are the answer. The test: "could I compute this from the draft alone, before publishing?" `MR-3.1` `MR-3.2`

Never outputs: a viral flag, a reach forecast, "2.4x", any absolute number. Bands and odds only, three reasons, one suggested action. `MR-1.2` `MR-8.10`

ANALOGY: GRADING ON THE PAGE'S OWN CURVE

Line the page's last 12 months of posts worst to best by spread: that is the **ladder**. A draft is scored by where it would likely land: bottom third for THIS page = at risk. A teacher grading within one class, never across schools (page "strong" lines differ 10-20x). The flop warning is a smoke alarm tuned to scream only at real fire: right 3 times in 4 or it stays silent, because an alarm that screams at toast gets its batteries pulled.

03 Exact steps to complete it (repo touch points verified)

1. **Lock D0** (the target): declare the pass line first, run candidates A + B through the VD-015B A/B harness on the time split, pick once, write into VD-PM-008 **RAFEH + ALEX · DAY 2**
2. **Reconcile live schema** (`SVV_COLUMNS`; repo DDL is ~12 columns stale) → build the ladder view (`thor_rdl_page_post_ladder.sql`, PERCENT_RANK within page, era-aware) → the feature view (FUSION + video + NLP + page state, as-of-date) **MUTEEB · WK 1**
3. **Train + export three artifacts** to `models/draft_score_v1/`: `model.txt`, isotonic calibrator, PCA + feature-spec manifest naming exact feature order **FAHEEM · WK 1-2**
4. **Serve**: clone the Fusion lambda (`marvel/DraftScore/`); CLIP-encode the draft in-process (parity gate: cosine > 0.999 vs the gold `.npy`); LightGBM + `pred_contrib` + `np.interp`; band vs the ladder passed by the app backend; provisioned concurrency 1-2 **MUTEEB + FAHEEM**
5. **Log from the first render**: `rdl.draft_predictions` + `draft_prediction_events` (draft, model version, advice shown, edits, outcome), parameterised inserts inside the idempotency wrapper; propose the 5-10% advice-free holdout for sign-off **MUTEEB · LAUNCH BLOCKER**
6. **Gate before ship**: beats caption-only · flop precision ≥75% · calibration near diagonal · band stable under small edits · sample discipline · everything logged (benchmarking protocol, page 7) **FAHEEM + RAFEH**

04 Best Times: same data, different machinery, separate surface

Best Times and the performance model read the **same warehouse** but are different kinds of claim. The model predicts an unpublished draft; Best Times describes when this page's audience historically responds. Keeping them apart is a product rule, not an accident: **timing advice never appears on the prediction surfaces** `MR-7.6`, Best Times is cut order #1 if the date slips, and each has its own honesty machinery:

	PERFORMANCE MODEL (S8)	BEST TIMES (S9)
Question	"Will THIS draft underperform on THIS page?"	"WHEN does this page's audience respond?"
Machinery	One global LightGBM, page-relative label	Per-page slot aggregation + shrinkage; Prophet weekly seasonality; changepoint drift
Refresh	Monthly retrain (44 posts/month: weekly content rules would be noise)	Weekly (day × hour cells accumulate samples fast) `MR-7.3`
Honesty guard	75% flop precision floor; bands + reasons; low-confidence states	Per-slot normalisation (one viral 3am post cannot paint a hot cell); cells under 5 posts silent; pooled day-part cells at our volume; heatmaps expire
Evidence	Flop band lands 22nd vs 42nd percentile (real, honest, weak-but-concentrated signal)	Personalised schedules beat global defaults, up to 17% reaction gain on Facebook (the ceiling, not the promise)

05 Contingency: we cannot get the topic trend score

Decision: ship v1 WITHOUT the World feature group. Planned, not a setback.
The trend badge needed an external ingestion pipeline (Wikipedia pageviews / Google Trends) that is not scoped `MR-7.5`. Three reasons this costs almost nothing: **(1) the measured value is small**: the one study that added exogenous trend signals to a virality model gained only +0.015 AUC-PR over text-only (ViralityNET 2026); **(2) nothing depends on it**: it was a display badge + one optional feature, never an input to bands, odds or reasons; **(3) LightGBM tolerates its absence natively** (missing-value handling), so if the pipeline arrives later it enters the monthly retrain as one new feature + one badge, no rework.

What replaces the "winner-side signal at draft time"?

Honestly: nothing fully does, and that is consistent with the physics (the top end is post-publish luck). The partial substitutes we DO ship: **upside odds** (historical frequency of 2x+ outcomes for this format/slot on this page) and, from S9, the **day-2 maturity grader**: observed early performance is worth more than any draft-time signal, trend scores included.

If a trend source lands later (fast-follow)

Requirements before it enters the model: free or fixed-cost source (Wikipedia pageviews qualifies), topic-matching via existing NLP topic labels, backfillable history for honest time-split evaluation, and the same pre-declared A/B protocol as every feature: no lift on the time split, no seat in the model.

06 Edge cases: the world changes under us

CASE	WHAT COULD GO WRONG	EXACT HANDLING
Meme culture / trends shift	The model learned last season's taste; patterns quietly stop working	Monthly retrain relearns it; pattern tables refresh monthly; the watchdog (S9) alerts on drift + quality-drop TOGETHER, so a taste shift the model survives does not page anyone; a shift it does not survive trips the retrain flag
Our own advice changes behaviour	Users kill flagged drafts → the flop base rate collapses → precision floor becomes unmeasurable → retraining learns from our own influence	Day-one logging of draft + advice + edits + outcome `MR-9.1`; intervention-aware retrain `MR-9.3`; proposed 5-10% advice-free holdout keeps an unconfounded stream (performative-prediction literature)
Meta kills another metric	15 Jun happens again	Target candidate A needs **no denominator** (shares percentile), so the label survives; the DEPRECATED_METRICS null-stub keeps schemas stable; the new failed-day guard makes the outage LOUD (day marked failed, nothing uploaded, watermark retries); quarterly single-metric probe added to the ops checklist
Engagement volumes shrinking platform-wide	VD-015B flagged month-over-month shrink; a 12-month ladder mixes fat months with lean ones, inflating the bar for new posts	Within-page percentile absorbs slow decline; the 90-day short ruler (current form) cross-checks the 12-month long ruler; trend_slope carries form into the score; if the gap widens, bands cut on the short ruler are the documented fallback
Seasonality (holidays, events)	December pages look "hot", January looks like decline; Prophet overfits changepoints on short series	Weekday/hour features in the model; Prophet fits weekly seasonality only (yearly needs 2+ years of history); changepoint priors kept tight on short series; the watchdog's joint rule stops seasonal swings paging anyone
The era break itself	Pre/post 15-Jun values are different units; naive stats read the break as a crash	Structural, never statistical: era flag on every row; no percentile ever pools across the break; forced changepoint at the boundary so the drift detector is architecturally incapable of flagging Meta's change as page behaviour; calibration refit post-break
Model rots silently	No labels arrive in production; accuracy decays invisibly	The S9 watchdog: KS tests on confidence distributions vs training baselines + no-label accuracy estimation (ATC), stratified human/DeepSeek spot-checks, healthy / watch / slipping states on the dashboard, documented trip action
Prophet fails on a page	Noisy small series produce garbage seasonality	Pre-declared gate: each page's Prophet fit must beat a seasonal-naive baseline on rolling-origin error or the page ships the simpler decomposition; log1p + winsorise before any fit so one viral post cannot bend the curve

THE ANALOGY: WEATHER, NOT PROPHECY

We run a weather station, not an oracle. Instruments recalibrate monthly (retrain), a second thermometer cross-checks the first (short vs long ruler), a technician watches the instruments themselves (watchdog), and when the meter gets swapped (era break) every reading is stamped with which meter took it. Nobody promises tomorrow's weather; we say "70% chance of rain" and we make sure that when we say 70%, it rains 7 times in 10.

07 Edge cases: individual pages and drafts

CASE	WHAT COULD GO WRONG	EXACT HANDLING
New page, thin history	No ladder to rank against; garbage confidence	Under 100 posts: confidence = low, said out loud `MR-1.7`. The global model still scores (it learned across pages), but page-state features shrink toward priors and no rule table reports until a slot has 5+ samples `MR-2.5`
Draft with too little signal	A bare one-word caption, no image: scoring it would be a guess	Below 5 usable signals: "not enough to score yet", never a fake number `MR-2.6`. Open: the exact definition of a usable signal `MR-2.7` (Faheem + Asad, week 1)
Page changes character (rebrand)	The ladder and heatmap describe the old audience	The trailing window adapts within months by construction; the drift detector + "rhythm changed, re-learning" state cover the gap; trend_slope tells the model the page is moving
One mega-viral outlier	A 13.8%-share freak bends every average it touches	Percentile labels are robust to outliers by construction; per-slot normalisation in timing; log1p + winsorise before decomposition; bounded-loss changepoint detection. The outlier keeps its top-10% tag on the answer sheet: history is allowed to be lucky
Video drafts	~46 videos total; no honest video ladder exists	Video is scored as a post with hook/scene as features `MR-5.2`; low confidence by construction `MR-5.5`; it never silently borrows an image-derived band. Video capability grows only when video volume exists `MR-5.6`
Carousels / multi-image	Which image is "the" content?	Classify + encode on the first/dominant image `MR-4.3`; multi-image fusion is post-MVP, documented
Boosted / paid posts	Paid reach contaminates the organic ladder, and Meta removed the paid/organic split (no way to filter)	Accepted, documented risk: within-page percentiles + robust stats dampen it; Lewis's domain read flags which pilots boost; if a pilot boosts heavily, their ladder honestly reflects their real practice: the product grades the page as it is actually run
FUSION backfill incomplete at training time	Image features missing for chunks of the window	Backfill is the critical path `MR-3.6` (newest 12 months first); where gaps remain, LightGBM's native missing-value handling + a has_fusion coverage flag; never silent zero-fill
Low-precision FUSION label	A shaky "meme" label becomes a client-facing reason	Labels below their per-label precision bar may enter the model as features but never appear in a sentence the user reads `MR-4.5`
Same-day multiple posts	Posts cannibalise each other's audience	gap-since-last-post is a feature; deeper cannibalisation modelling deliberately deferred (documented, not forgotten)
Deleted posts / missing outcomes	Logged predictions with no answer sheet	Outcome-join at grading time skips them from training and records the deletion as an event in `draft_prediction_events`: absence of outcome is itself data
Art-E asked about dead metrics	Hallucinated impressions numbers	Refusal with the reason ("Meta removed that metric in June 2026"), seeded in the 200-question gold set; metric enums in tool schemas simply do not contain dead names
Prompt injection via comments	Any commenter can write "ignore previous instructions" into text Art-E retrieves	Retrieved rows are data, never instructions: JSON-encoded inside tool results, untrusted-content policy in the system prompt, red-team gate with seeded attacks before the demo

08 Benchmarking: the three rulers, then three rings of proof

RULER	WHAT IT MEASURES	REBUILT	NEEDS
Long ruler	The page's own trailing 12 months: median, strong line, top-10% line. Grades history AND feeds the model as page state	Monthly	30+ posts, else "not enough history"
Short ruler	Same lines on the last 90 days = current form. The gap between short and long = warming / cooling (trend_slope)	Monthly	Cross-checks the long ruler when volumes shift
Peer ruler	Archetype-group comparison ("similar pages run 3.1%, you run 2.4%"): arrives with the S9 cross-page graph	With graph	5+ peer pages; cross-page claims stay inputs until they hold on the page's own ladder `MR-2.9`

Ring 1 · Offline, before anything ships

Strict time split (train past, test future, page-stratified; random splits banned `MR-10.1`). **Baseline to beat: caption-only**: tying it means the model learned nothing from the image `MR-10.2`. Metrics that matter: rank correlation, flop-band separation (the 22-vs-42 percentile gap), calibrated precision at the flop flag, per-era. **Pass lines are committed to the repo before results are seen** (pre-registration): the same discipline that caught the NLP zero-lift honestly in VD-015B.

Ring 2 · Gates at the door

Flop precision ≥75% validated or the flag does not fire `MR-8.3` · calibration curve near diagonal (80% means 4-in-5) `MR-8.4` · band stability under small edits (jitter gate, VD-QA-AI-008) `MR-8.5` · no slot under 5 samples, no score under min-signal `MR-10.4` · Art-E gold set: accuracy, citation recall + precision, refusal correctness, injection resistance, with a citation-precision floor as the twin of the flop floor · Best Times (S9): held-out-week lift vs the pre-declared line, within page, content-controlled, with placebo checks (permuted slots and stale heatmaps must show NO lift).

Ring 3 · Online, forever

Every prediction logged with the advice shown `MR-9.1` · monthly **predicted-vs-actual receipts**, on the record for any pilot who asks `MR-9.4` · the advice-free holdout keeps metrics measurable after our own warnings change behaviour · the watchdog (S9) monitors the models' own output distributions so rot is caught between retrains · Rafeh's weekly Art-E answer reviews against the written bar.

The non-engineer test (the last gate)

Per VD-PM-008 and `MR-10.5`: could a media buyer read the chip and know what to do? Act on the warning without asking what it means? Check a citation themselves? If acceptance needs an engineer in the room, it fails. Rafeh runs it on real pilot data at sprint end; evidence attached, pass / fail / defer recorded.

ANALOGY: THE EXAM BOARD

Ring 1 is marking the mock exam under exam conditions (no peeking: the time split). Ring 2 is the exam board refusing to publish grades that fail moderation (the gates). Ring 3 is checking, every month, that last year's grades still predict how students actually did (receipts). And the pass mark is agreed **before** anyone sits the paper: that is what keeps us honest with ourselves.

09 The Model Master Rules: a constitution, not a spec

The Master Rules document (24 Jul, built on Rafeh's VD-PM-008 product definitions) is the **rules layer**: it says what the model must and must not do; the specs say how. Where a spec and the rules disagree, the spec wins and the disagreement is logged in its conflicts table. Its power is in three mechanisms:

1 · Rules are citable, so decisions stop being re-argued

Every rule has an id (`MR-3.1` = the leakage ban, `MR-8.3` = the precision floor). Tickets, QA gates, code reviews and this document cite ids instead of re-litigating. When a reviewer writes "this feature violates MR-3.2", the argument is already over. Both S8 test suites (VD-QA-AI-008, VD-QA-008) map their gates to MR-8.x and MR-10.x line by line.

2 · Confidence tags say what is actually buildable

Every rule carries a tag: **[C]** confirmed in a locked source · **[X]** constructed (inferred, unreviewed) · **[P]** provisional (a real decision, unmade) · **[G]** gap (no decision exists: do not build to it). This is rare honesty in internal docs: an engineer can see at a glance that the band cut method is locked but the numeric cut is [P], and that the whole sensitivity taxonomy (§6) is [G]: structure only, build nothing.

3 · The conflicts table forces decisions into the open

§11 lists 7 contradictions between in-force documents, each with an owner. Conflict #1 (which target formula) is exactly what the Meta deprecation then detonated: because it was already logged with owners, the escalation could say "this forces Conflict #1 closed" instead of discovering the ambiguity mid-incident. Conflict #3 (flop = bottom third vs bottom quarter) is D2 on the sprint board with Rafeh's name on it.

4 · What it needs from us this sprint

The document is provisional and unreviewed by its own admission. S8 owes it: the Alex + Rafeh + Faheem review pass · D0 and D2 resolved and written back · the usable-signal definition (`MR-2.7`) · Filza's refusal-wording sign-off (`MR-8.7`) · and each [P]/[G] either promoted to [C] or explicitly deferred. The rules doc is only as strong as its last reconciliation.

WHERE A RULE BITES IN S8	THE RULE	WHAT IT PREVENTS
Feature engineering	`MR-3.1` / `MR-3.2` leakage ban + presumption	An accidentally-brilliant model that peeked at the answer
Product copy	`MR-1.2` bands not numbers · `MR-8.10` one action	"This will hit 13.8%" appearing anywhere, including Art-E free text
The warning threshold	`MR-8.3` tune the threshold, never the promise	Quietly lowering the bar to make the flag fire more
Cross-page features	`MR-2.9` learned across, delivered per page	"Pages like yours" claims without 5+ samples on their own ladder
Narrative	`MR-8.9` the firewall: internally honest, externally "audience engagement insights"	Meta App Review reading "virality prediction engine"

10 The papers, by question they answer

CEILING **Exploring limits to prediction in complex social systems** · Martin, Hofman, Watts et al, WWW 2016 · even unlimited data explains under half of success · arxiv.org/abs/1602.01013

CEILING **Early Multimodal Prediction of Meme Virality** · Dogan et al, 2025 · content-only PR AUC ~0.13; 30 min of live engagement → ROC AUC 0.92 · arxiv.org/abs/2510.05761

REFRAME **Can Cascades be Predicted?** · Cheng, Adamic, Kleinberg, Leskovec, WWW 2014 · balanced band targets work on Facebook photos (0.795) where absolute prediction fails · arxiv.org/abs/1403.4608

TARGET **Beyond Views: relative engagement** · Wu, Rizoiu, Xie, ICWSM 2018 · percentile-normalised engagement is the stable, predictable quantity (R^2 0.77) · arxiv.org/abs/1709.02541

TARGET **ViralityNet (trend cross-attention)** · 2026 · per-community percentile + absolute floor is the standard target; exogenous trend signals added only +0.015 AUC-PR · arxiv.org/abs/2605.02358

TARGET **The Structural Virality of Online Diffusion** · Goel, Anderson, Hofman, Watts, Mgmt Sci 2016 · a billion cascades measured from shares alone, no impression data · 5harad.com/papers/twiral.pdf

TARGET **Cats and Captions vs Creators and the Clock** · Hessel, Lee, Mimno, WWW 2017 · content signal emerges only under within-community, timing-controlled labels · arxiv.org/abs/1703.01725

DRIFT **Learning under Concept Drift: A Review** · Lu et al, TKDE 2019 · era-stratify, never pool across a definition break, refit calibration · arxiv.org/abs/2004.05785

MODEL **Why do tree-based models still outperform deep learning on tabular data?** · Grinsztajn et al, NeurIPS 2022 · our data regime is GBDT territory · arxiv.org/abs/2207.08815 · plus McElfresh 2023: tuning beats architecture swaps · arxiv.org/abs/2305.02997

MODEL **SMP Challenge overview + 2023-25 winners** · every podium is GBDT over frozen embeddings; the 2024 winner's decisive signal was account-level history (our page state) · arxiv.org/abs/2405.10497, arxiv.org/abs/2507.00926, arxiv.org/abs/2507.00950

CALIB **Predicting Good Probabilities** · Niculescu-Mizil, Caruana, ICML 2005 · isotonic safe only above ~1,000 points (else Platt); boosted trees need calibration · cs.cornell.edu/~alexn/papers/calibration.icml05.crc.rev3.pdf · isotonic via PAV: Zadrozny, Elkan, KDD 2002 · cseweb.ucsd.edu/~elkan/kdd2002.pdf

REASONS **TreeExplainer (exact SHAP for trees)** · Lundberg et al, Nature MI 2020 · exact additive per-prediction reasons, cheap at serve time · arxiv.org/abs/1905.04610 · foundation: Lundberg, Lee, NeurIPS 2017 · arxiv.org/abs/1705.07874

IMAGE **Dissecting the Meme Magic** · Ling et al, CSCW 2021 · visual composition separates shared from unshared memes (AUC 0.866) · arxiv.org/abs/2101.06535 · honest ceiling: visuals add ~+0.009 rank-corr once page + timing are in (Riis 2021) · arxiv.org/abs/2004.12482

DUEL **When is it Better to Compare than to Score?** · Shah et al, 2014 · pairwise formally beats absolute scoring under high noise: the Draft Duel license · arxiv.org/abs/1406.6618

LOOP **Performative Prediction** · Perdomo et al, ICML 2020 · predictions change the distribution they predict · arxiv.org/abs/2002.06673 · logging blueprint: Bottou et al, JMLR 2013 · arxiv.org/abs/1209.2355 · feedback-loop degradation: Chaney 2018 · arxiv.org/abs/1710.11214

TIMING **When-To-Post on Social Networks** · Spasojevic et al, KDD 2015 · personalised schedules: up to 17% reaction gain on Facebook (the ceiling) · arxiv.org/abs/1506.02089 · timing is feed competition, not clock magic: RedQueen, WSDM 2017 · arxiv.org/abs/1610.05773

VELOCITY **Predicting popularity from early patterns** · Szabo, Huberman 2008 · arxiv.org/abs/0811.0405 · daily-grain works with delta vectors: Pinto et al, WSDM 2013 (via the Tatar survey) · the honest basis of the day-2 grader

WATCHDOG **Failing Loudly** · Rabanser et al, NeurIPS 2019 · the best shift detector is KS tests on the model's own outputs · arxiv.org/abs/1810.11953 · no-label accuracy: ATC, Garg et al, ICLR 2022 · arxiv.org/abs/2201.04234 · spot-check design: Kossen et al, ICML 2021 · arxiv.org/abs/2103.05331

ART-E **BIRD + Spider 2.0** · free-form text-to-SQL: ~82% realistic / ~21% enterprise = disqualifying · arxiv.org/abs/2305.03111, arxiv.org/abs/2411.07763 · semantic layer 3.4x: Sequeda 2023 · arxiv.org/abs/2311.07509 · no in-context arithmetic: TableBench · arxiv.org/abs/2408.09174

ART-E **ALCE citations benchmark** · Gao et al, EMNLP 2023 · citation recall + precision metrics · arxiv.org/abs/2305.14627 · deployed engines audit at 51.5% supported: Liu et al 2023 · arxiv.org/abs/2304.09848 · refusal must be trained + measured: R-Tuning · arxiv.org/abs/2311.09677

SECURITY **Indirect prompt injection** · Greshake et al, CCS 2023 · arxiv.org/abs/2302.12173 · spotlighting defence cuts attack success >50% → <2%: Hines et al 2024 · arxiv.org/abs/2403.14720 · LLM-judge caveats: Zheng et al, NeurIPS 2023 · arxiv.org/abs/2306.05685

Note: paperswithcode.com was sunset in July 2025 (it now redirects to Hugging Face Papers); all links above were fetched directly from arXiv / publisher pages and verified against their abstracts on 28 Jul 2026. The full 58-paper set with per-paper application notes lives on the Sprint Playbook page (S8 + S9 tabs).

11 Definition of done for the strategy (not just the code)

- ☐ **The model map is agreed:** one LightGBM for scoring, tables for patterns, Prophet for rhythm, no oracle: and nobody on the team can be surprised by which is which
- ☐ **D0 + D2 locked and written into VD-PM-008** (target formula; flop cut third vs quarter), pass lines pre-declared before any validation runs
- ☐ **The World feature group formally marked optional**, trend badge moved to fast-follow, the contingency on page 4 pasted into the tracker ticket
- ☐ **Every edge case on pages 5-6 has an owner behaviour in the shipped product** (honest states, era stamps, low-confidence paths), not just a line in this document
- ☐ **All three benchmarking rings live:** time-split harness (Ring 1), the gates wired into VD-QA-AI-008 / VD-QA-008 (Ring 2), logging + receipts + holdout proposal decided (Ring 3)
- ☐ **The Master Rules review pass happens:** Alex + Rafeh + Faheem sign the [C] rules, resolve or defer every [P]/[G] touched by S8
- ☐ **The incident closes properly:** PR #7 merges with the guard, the ALTERs run, the empty partitions are deleted, rows land past 13 Jun

12 Open decisions, named

#	DECISION	OWNER	DUE
D0	Target formula: log(shares) percentile + floor (A) vs shares-per-viewer (B), via the pre-declared A/B	Rafeh + Alex (Faheem's pull)	Day 2
D1	Draft-score + retrieval interface contracts	Faheem + Muteeb + Asad	Day 1
D2	Flop cut: bottom third vs bottom quarter (Master Rules conflict #3)	Rafeh	Day 2
D3	"Usable signal" definition (`MR-2.7`) + Art-E refusal wording sign-off (`MR-8.7`)	Faheem + Asad · Filza	Week 1
D4	The 5-10% advice-free logging holdout: adopt or decline (research recommendation)	Rafeh + Asad	Week 1
D5	Master Rules review: promote / defer every [P] and [G] that S8 touches	Alex + Rafeh + Faheem	Sprint end

THE WHOLE STRATEGY IN THREE SENTENCES

We predict what is predictable (the bottom of the ladder), we look up what is countable (the pattern tables), and we refuse to prophesy what is luck (the top). Everything the user sees carries its reasons, its odds and its receipt, and everything we ship is benchmarked against a pass line we wrote down before we saw the results. When the world changes, the machinery notices before the users do.

COMPANION DOCUMENTS: SPRINT 8 BUILD INSTRUCTION (THE HOW) · SPRINT PLAYBOOK S8 + S9 TABS (LIVING VERSION) · MODEL MASTER RULES 24 JUL (THE CONSTITUTION) · MARVEL PR #7 (THE INCIDENT FIX, IN REVIEW)